Vectors I: Homework 3

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Vector product: properties, parallel vectors, areas, collinearity via cross product. Q1 True/False from the Basics Worksheet (Vector Product). Q2–Q7 are full promotional-examination questions.

1

For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\times\mathbf{a}=\mathbf{0}$.	
(b)	$\mathbf{a}\times\mathbf{a}=0$.	
(c)	$\mathbf{i}\times\mathbf{j}=\mathbf{k}$.	
(d)	$\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$.	
(e)	$\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$.	
(f)	$\mathbf{a}\times\mathbf{b}$ is a scalar.	
(g)	$\mathbf{k}\times\mathbf{j}=\mathbf{i}$.	
(h)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$.	
(i)	$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$.	
(j)	$(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$.	
(k)	$\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.	
(l)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar.	
(m)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.	
(n)	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(o)	If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.	
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$.	
(q)	The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $\lvert\mathbf{a}\times\mathbf{b}\rvert$.	
(r)	The area of triangle OAB is $\lvert\overrightarrow{OA}\times\overrightarrow{OB}\rvert$.	
(s)	$\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector.	
(t)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$.	
(u)	$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.	
(v)	$\lvert\mathbf{a}\times\mathbf{b}\rvert^2+(\mathbf{a}\cdot\mathbf{b})^2=\lvert\mathbf{a}\rvert^2\lvert\mathbf{b}\rvert^2$.	
(w)	$\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.	
(x)	$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$.	
(y)	The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $\lvert\overrightarrow{AB}\times\mathbf{d}\rvert$.	

Questions

2 [JPJC 2024 Promo Q11]

Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors. Point C is the midpoint of OA and point D lies on BC such that $BD:DC=2:3$.

(i) Find the position vectors $\overrightarrow{OC}$ and $\overrightarrow{OD}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. **[2]**

(ii) Find the area of triangle OCD in the form of $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be found. **[3]**

(iii) Given that $|\mathbf{a}|=2$, $|\mathbf{b}|=\sqrt{2}$ and angle AOB is $\frac{\pi}{4}$ radian, using scalar product, show that CD is perpendicular to OA. **[3]**

(iv) Given instead that $\mathbf{a}$ is a unit vector, give a geometrical interpretation of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

3 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC:AB=1:3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. **[2]**

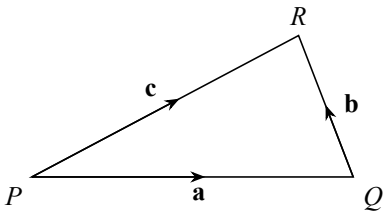
(ii) Show that the area of triangle OAC is given by $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be determined. **[2]**

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. **[3]**

4 [ASRJC 2024 Promo Q5]

In triangle PQR, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ represent the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{PR}$ respectively.



(a) State a geometrical interpretation of $|\mathbf{c}\times\hat{\mathbf{a}}|$ and hence show that the area of the triangle PQR can be written as $\frac{1}{2}|\mathbf{c}\times\mathbf{a}|$ units^2. **[3]**

(b) With clear explanations, show that $|\mathbf{c}\times\mathbf{a}|=|\mathbf{b}\times\mathbf{a}|$ and deduce that

$$\frac{|\mathbf{c}|}{\sin\angle PQR}=\frac{|\mathbf{b}|}{\sin\angle QPR} \quad \text{(Sine Rule)}.$$

[2]

5 [CJC 2024 Promo Q7]

(a) Three non-zero vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are such that $3\mathbf{p}\times\mathbf{q}=\mathbf{r}\times\mathbf{p}$.
Find a linear relationship between $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$. **[3]**

(b) Referred to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$.
It is given that $\mathbf{a}$ and $\mathbf{b}$ are non-parallel unit vectors.
The point B divides AC in the ratio $1:3$.
The point D lies on OB produced such that $OB:OD=1:m$, where $m\in\mathbb{Z}$, $m\geq 2$.
Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, find the numerical value of m. **[4]**

6 [SAJC 2024 Promo Q9]

(a) Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

(i) By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. **[3]**

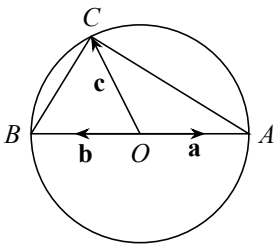
(ii) Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

(iii) The points P, Q and R have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of λ. **[3]**

(b) In the triangle OUV where O is the origin, the position vectors of the points U and V are $\mathbf{u}$ and $\mathbf{v}$ respectively. The point W is the midpoint of OU, and the point X has position vector given by $\frac{1}{16}\mathbf{u}+\frac{3}{4}\mathbf{v}$. Show that the area of triangle UWX can be written as $m|\mathbf{u}\times\mathbf{v}|$ where m is a constant to be found. **[3]**

7 [CJC 2025 Promo Q6]

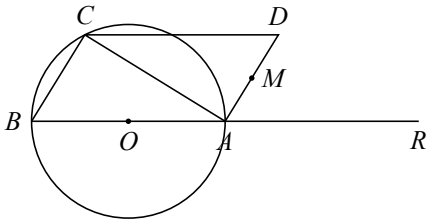
The diagram shows a unit circle with centre O and diameter AB. The point C lies on the circumference of the circle. The position vectors of A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively.



(a) By considering an appropriate scalar product, show that AC and BC are perpendicular. **[4]**

(b) Give a geometrical interpretation of $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$. **[1]**

The point D is such that $ABCD$ is a parallelogram. The point M is the mid-point of AD and the point R lies on OA produced such that $OA:AR=1:2$.



(c) By considering $\overrightarrow{CM}\times\overrightarrow{CR}$ or otherwise, show that the points C, M and R are collinear. **[3]**

Answers

1 **(a)** T **(b)** F **(c)** T **(d)** F **(e)** T **(f)** F **(g)** F **(h)** T **(i)** T **(j)** T **(k)** T **(l)** T **(m)** T **(n)** F **(o)** F **(p)** T **(q)** T **(r)** F **(s)** F **(t)** T **(u)** T **(v)** T **(w)** F **(x)** F **(y)** F

2 **(i)** $\overrightarrow{OC}=\frac{1}{2}\mathbf{a}$, $\overrightarrow{OD}=\frac{1}{5}(\mathbf{a}+3\mathbf{b})$

(ii) $\frac{3}{20}|\mathbf{a}\times\mathbf{b}|$, $k=\frac{3}{20}$

(iii) Shown.

(iv) $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$ (or a line or vector parallel to $\overrightarrow{OA}$).

3 **(i)** $\mathbf{c}=\frac{2\mathbf{a}+\mathbf{b}}{3}$

(ii) Shown. $k=\frac{1}{6}$

(iii) 1

4 **(a)** $|\mathbf{c}\times\hat{\mathbf{a}}|$ represents the perpendicular distance from the point R to the line PQ. Shown.

(b) Shown.

5 **(a)** $\mathbf{p}=k(3\mathbf{q}+\mathbf{r})$, $k\in\mathbb{R}\setminus\{0\}$

(b) $m=7$

6 **(a)(i)** Shown.

(a)(ii) $|\mathbf{a}\cdot\mathbf{b}|$ is the length of projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$.

(a)(iii) $\lambda=-25$

(b) $m=\frac{3}{16}$

7 **(a)** Shown.

(b) $|\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})|$ is the length of projection of $\overrightarrow{AC}$ onto $\overrightarrow{OA}$.

(c) Shown.

END OF WORKSHEET